

Lab 05

5t OTA

PART 1: sizing

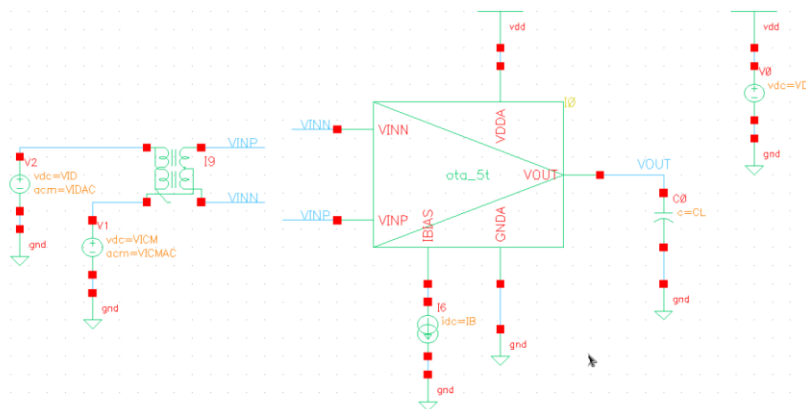
- 1) We want to design a 5t OTA with the parameters below.

Parameter	65nm CMOS
GBW	5 MHz
DC gain	34 dB
Supply	1.2V
OTA current consumption	20 μA
CM input range	0.6 – 0.9 V
IBIN	20 μA
Output load	5 pF

- 2) Choose an appropriate sizing to achieve the required specs.

PART 2: OTA Simulation

- 1) Create a testbench for the OTA, you can use the testbench below as a reference.



- 2) Report Schematic of the OTA with DC node voltages clearly annotated. Use VICM at the middle of the CMIR.
- 3) What is DC voltage at VOUT?

1. AC Analysis (part 1)

- 1) Set $VIDAC = 1$ and $VICMAC = 0$ and Use VICM at the middle of the CMIR.
- 2) Plot the gain vs frequency.
- 3) Report the Gain, BW and GBW.
- 4) Compare the results with hand analysis.

2. AC Analysis (part 2)

- 1) Set $VICMAC = 1$ and $VIDAC = 0$. Use VICM in the middle of the CMIR.
- 2) Plot CM gain vs frequency.
- 3) Compare the results with hand analysis.
- 4) Calculate the CMRR.